

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    "Order_ID": [101,102,103,104,105,106,107,108,109,110],
    "Order_Time": ["12:30 PM","01:15 PM","08:00 PM","07:30 PM","02:00 PM",
                  "09:00 PM","12:45 PM","08:30 PM","01:00 PM","07:45 PM"],
    "Location": ["Chennai","Madurai","Trichy","Chennai","Coimbatore",
                "Madurai","Chennai","Trichy","Coimbatore","Madurai"],
    "Food_Item": ["Pizza","Burger","Biryani","Pizza","Sandwich",
                 "Fried Rice","Biryani","Pizza","Burger","Noodles"],
    "Quantity": [2,1,3,1,2,1,2,1,2,1],
    "Price": [350,250,400,350,200,180,400,350,250,220]
}

df = pd.DataFrame(data)

df
```

	Order_ID	Order_Time	Location	Food_Item	Quantity	Price
0	101	12:30 PM	Chennai	Pizza	2	350
1	102	01:15 PM	Madurai	Burger	1	250
2	103	08:00 PM	Trichy	Biryani	3	400
3	104	07:30 PM	Chennai	Pizza	1	350
4	105	02:00 PM	Coimbatore	Sandwich	2	200
5	106	09:00 PM	Madurai	Fried Rice	1	180
6	107	12:45 PM	Chennai	Biryani	2	400
7	108	08:30 PM	Trichy	Pizza	1	350
8	109	01:00 PM	Coimbatore	Burger	2	250
9	110	07:45 PM	Madurai	Noodles	1	220

```
df["Total_Amount"] = df["Quantity"] * df["Price"]

df
```

	Order_ID	Order_Time	Location	Food_Item	Quantity	Price	Total_Amount
0	101	12:30 PM	Chennai	Pizza	2	350	700
1	102	01:15 PM	Madurai	Burger	1	250	250
2	103	08:00 PM	Trichy	Biryani	3	400	1200
3	104	07:30 PM	Chennai	Pizza	1	350	350
4	105	02:00 PM	Coimbatore	Sandwich	2	200	400
5	106	09:00 PM	Madurai	Fried Rice	1	180	180
6	107	12:45 PM	Chennai	Biryani	2	400	800
7	108	08:30 PM	Trichy	Pizza	1	350	350
8	109	01:00 PM	Coimbatore	Burger	2	250	500
9	110	07:45 PM	Madurai	Noodles	1	220	220

```
df.isnull().sum()
```

[Show hidden output](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order_ID        10 non-null    int64
1   Order_Time      10 non-null    object
2   Location         10 non-null    object
3   Food_Item       10 non-null    object
4   Quantity        10 non-null    int64
5   Price           10 non-null    int64
6   Total_Amount    10 non-null    int64
dtypes: int64(4), object(3)
memory usage: 692.0+ bytes
```

```
popular_dishes = df["Food_Item"].value_counts()
```

```
popular_dishes
```

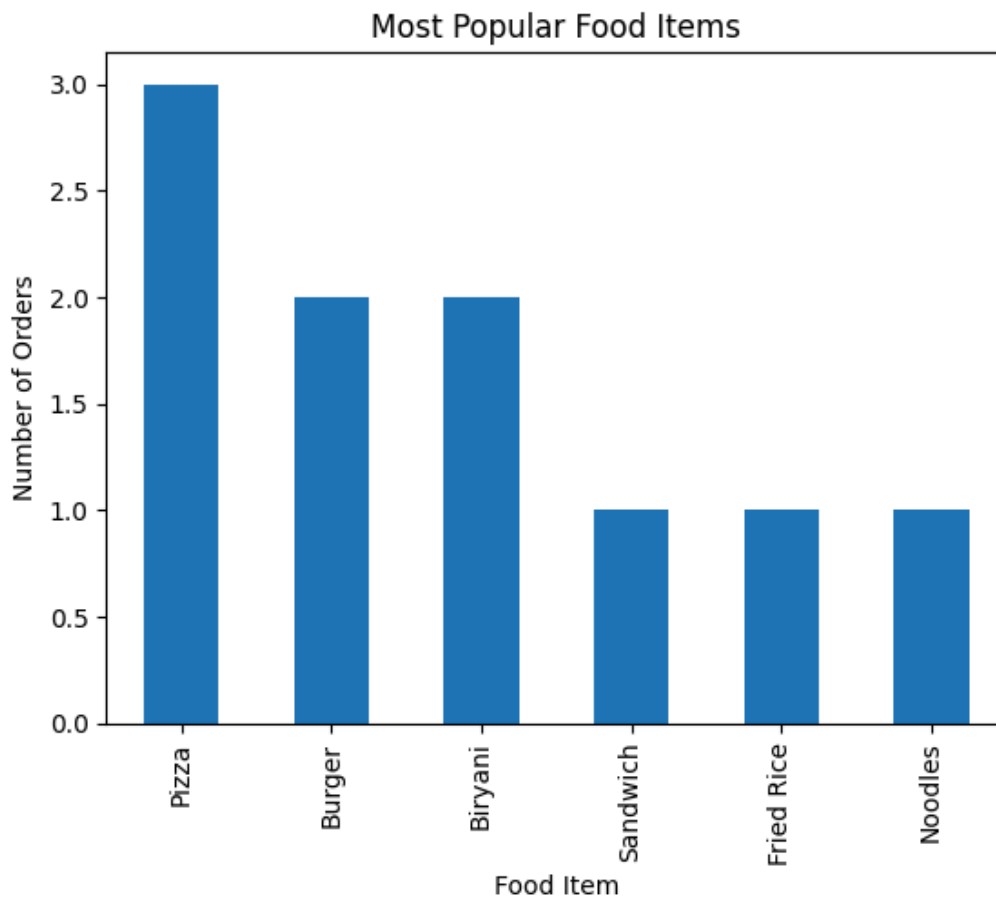
	count
Food_Item	
Pizza	3
Burger	2
Biryani	2
Sandwich	1
Fried Rice	1
Noodles	1

```
dtype: int64
```

```
popular_dishes.plot(kind="bar")

plt.title("Most Popular Food Items")
plt.xlabel("Food Item")
plt.ylabel("Number of Orders")

plt.show()
```



Start coding or [generate](#) with AI.

```
location_orders = df["Location"].value_counts()

location_orders
```

	count
Chennai	3
Madurai	3
Trichy	2
Coimbatore	2

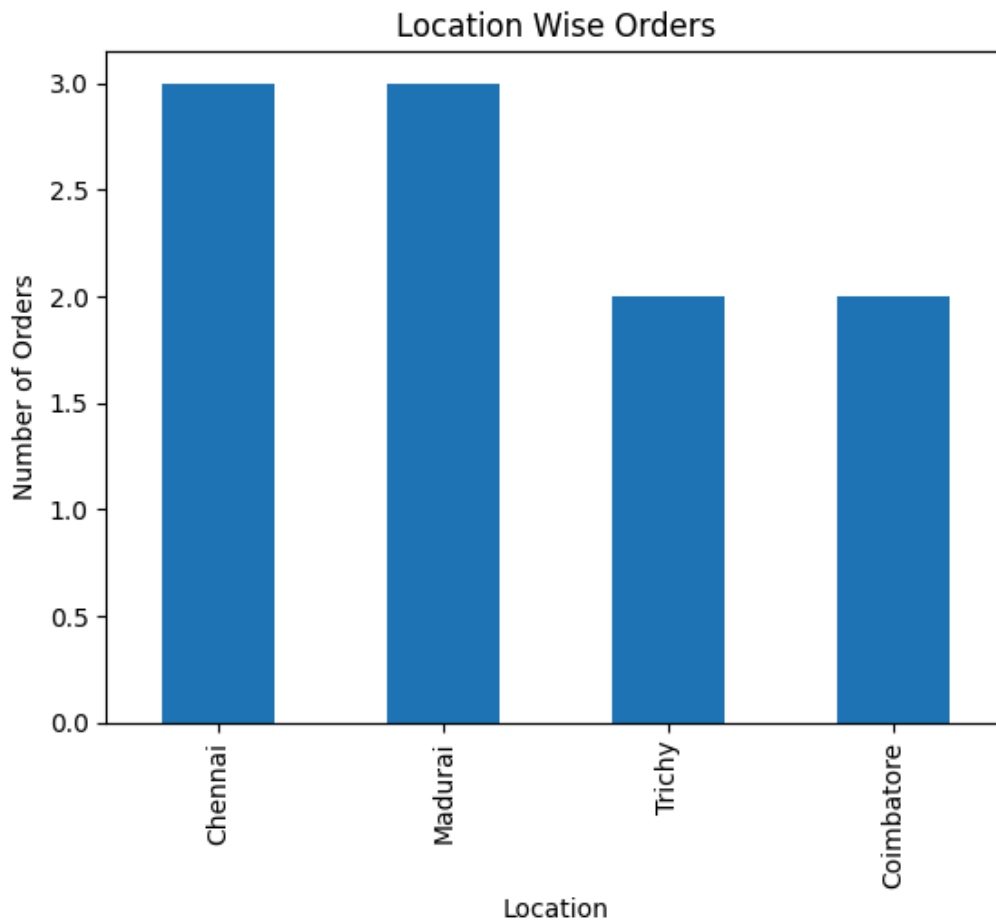
dtype: int64

```
location_orders.plot(kind="bar")

plt.title("Location Wise Orders")
```

```
plt.xlabel("Location")
plt.ylabel("Number of Orders")

plt.show()
```



```
df["Hour"] = pd.to_datetime(df["Order_Time"]).dt.hour
```

```
df[["Order_Time", "Hour"]]
```

```
/tmp/ipykernel_866/2413498755.py:1: UserWarning: Could not infer format, so each element
df["Hour"] = pd.to_datetime(df["Order_Time"]).dt.hour
```

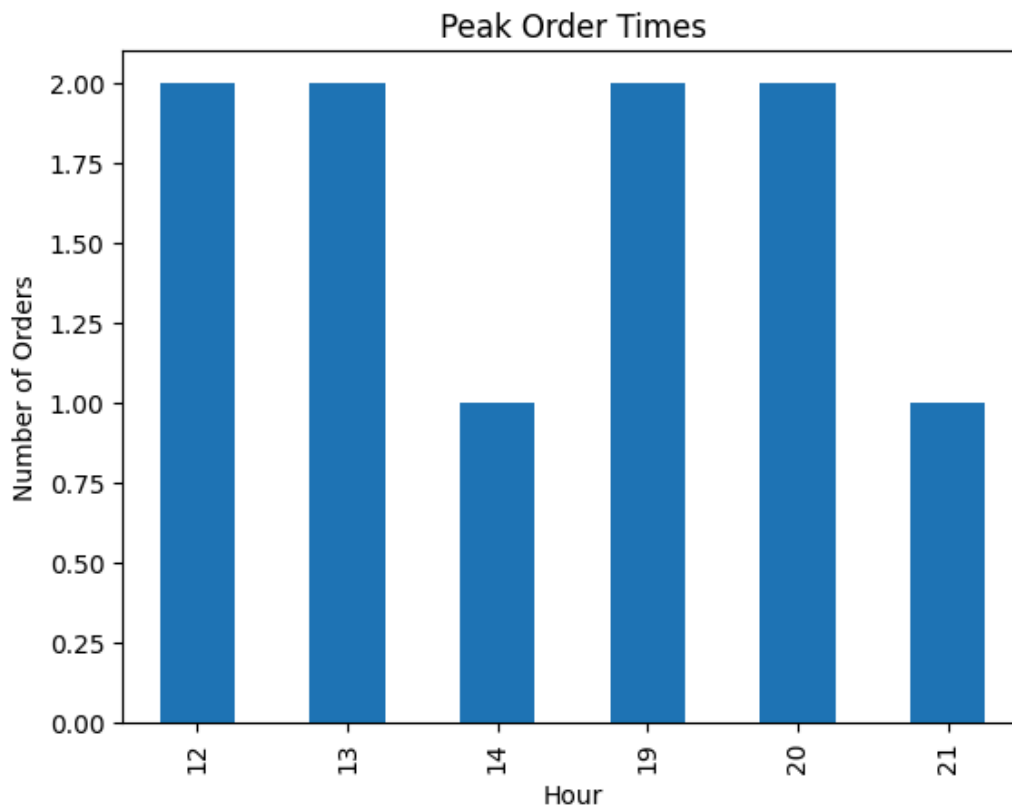
	Order_Time	Hour
0	12:30 PM	12
1	01:15 PM	13
2	08:00 PM	20
3	07:30 PM	19
4	02:00 PM	14
5	09:00 PM	21
6	12:45 PM	12
7	08:30 PM	20
8	01:00 PM	13
9	07:45 PM	19

```
peak_time = df["Hour"].value_counts().sort_index()

peak_time.plot(kind="bar")

plt.title("Peak Order Times")
plt.xlabel("Hour")
plt.ylabel("Number of Orders")

plt.show()
```



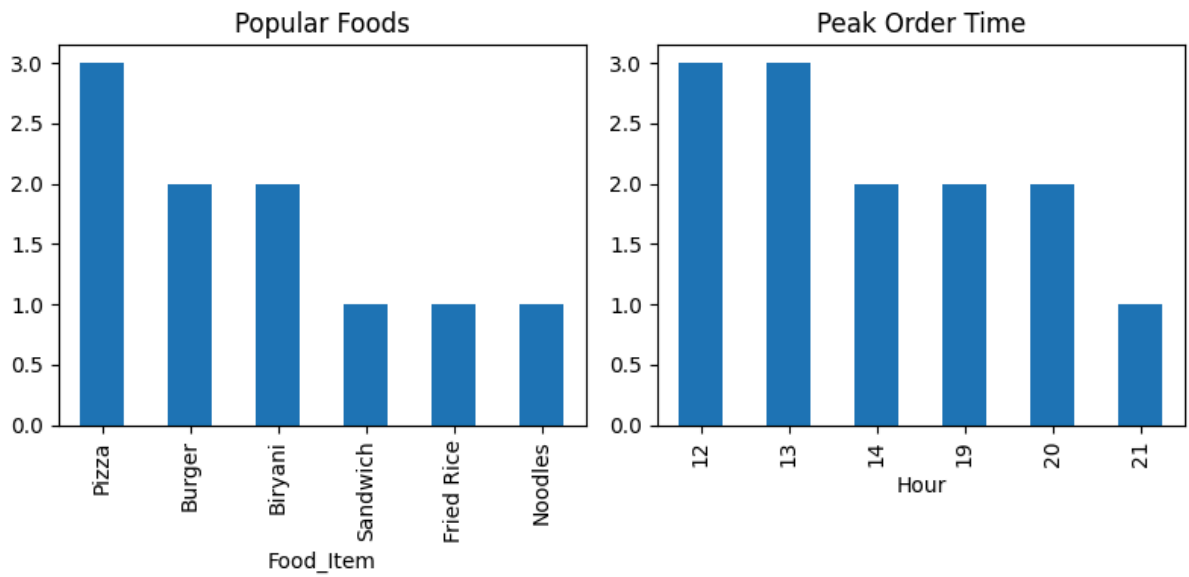
```
plt.figure(figsize=(12,4))

# Chart 1
plt.subplot(1,3,1)
df["Food_Item"].value_counts().plot(kind="bar")
plt.title("Popular Foods")

# Chart 2
plt.subplot(1,3,2)
df["Location"].value_counts().plot(kind="bar")
plt.title("Orders by Location")

# Chart 3
peak_time = df["Hour"].value_counts().sort_index()
peak_time.plot(kind="bar")
plt.title("Peak Order Time")

plt.tight_layout()
plt.show()
```



```
location_orders = df["Location"].value_counts()

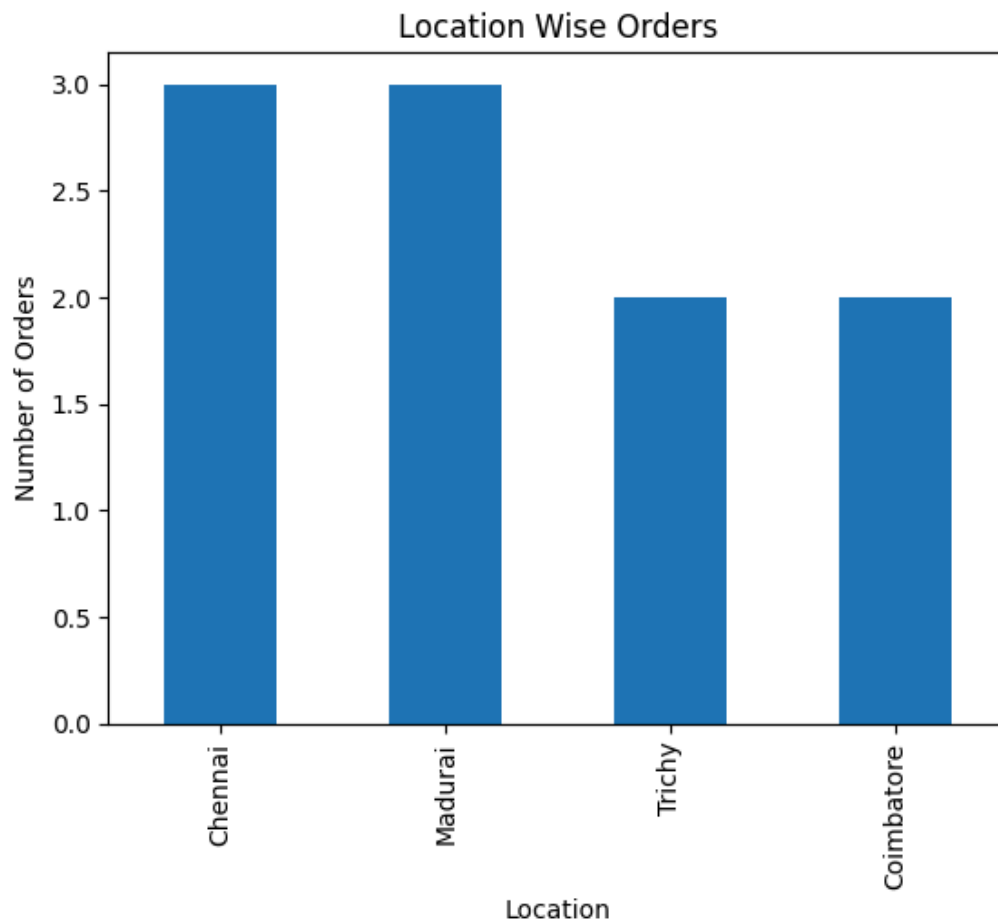
print(location_orders)
```

```
Location
Chennai      3
Madurai      3
Trichy       2
Coimbatore   2
Name: count, dtype: int64
```

```
location_orders.plot(kind="bar")

plt.title("Location Wise Orders")
plt.xlabel("Location")
plt.ylabel("Number of Orders")

plt.show()
```



```
sales_by_item = df.groupby("Food_Item")["Total_Amount"].sum()

sales_by_item.plot(kind="bar")

plt.title("Sales by Food Item")
plt.xlabel("Food Item")
plt.ylabel("Total Sales Amount")

plt.show()
```

Sales by Food Item



```
plt.figure(figsize=(14,4))

plt.subplot(1,3,1)
df["Food_Item"].value_counts().plot(kind="bar")
plt.title("Popular Foods")

plt.subplot(1,3,2)
df["Location"].value_counts().plot(kind="bar")
plt.title("Orders by Location")

plt.subplot(1,3,3)
df["Hour"].value_counts().sort_index().plot(kind="bar")
plt.title("Peak Order Time")

plt.tight_layout()
plt.show()
```

